

## MIDLAND INTL, TX, USA

WMO: 722650

Lat: 31.948N

Lon: 102.209W

Elev: 2862

StdP: 13.24

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 23023

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 20.7       | 24.6 | 0.8                           | 6.3 | 39.6 | 4.9 | 7.9 | 44.7 | 32.1                  | 54.3 | 27.8 | 53.0 | 8.4                   | 30   | 0.574 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 22.7                   | 101.6           | 67.2 | 99.2 | 67.4 | 97.0 | 67.5 | 73.1                | 88.5 | 72.2 | 87.5 | 71.3 | 86.6 | 12.5                 | 150  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 69.4                            | 120.4 | 76.3 | 68.4 | 115.9 | 76.3 | 67.4 | 112.0 | 76.4 | 38.5          | 88.6 | 37.7 | 87.5 | 37.0 | 86.5 | 77.4              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%   | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |
|                   |      |      | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
| 26.9              | 24.2 | 20.9 | 14.2                       | 106.2 | 3.5                | 3.4 | 11.7                                               | 108.6 | 9.7        | 110.6 | 7.7        | 112.5 | 5.1        | 115.0 |
|                   |      |      | 12.2                       | 74.5  | 3.2                | 1.3 | 10.0                                               | 75.5  | 8.1        | 76.2  | 6.3        | 76.9  | 4.0        | 77.9  |
|                   |      |      | DB                         |       |                    |     |                                                    |       |            |       |            |       |            |       |
|                   |      |      | WB                         |       |                    |     |                                                    |       |            |       |            |       |            |       |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 65.2   | 45.2  | 49.5  | 57.1  | 65.5  | 74.5  | 82.0  | 83.4  | 82.5  | 75.6  | 65.8  | 54.0  | 45.6  |
|                                                                        | DBStd       | 15.98  | 8.99  | 10.13 | 9.59  | 8.47  | 7.78  | 5.27  | 4.26  | 4.37  | 6.24  | 8.38  | 8.89  | 8.71  |
|                                                                        | HDD50       | 626    | 202   | 120   | 43    | 6     | 0     | 0     | 0     | 0     | 0     | 6     | 61    | 188   |
|                                                                        | HDD65       | 2475   | 615   | 437   | 274   | 95    | 17    | 0     | 0     | 0     | 6     | 92    | 338   | 601   |
|                                                                        | CDD50       | 6156   | 52    | 108   | 263   | 472   | 760   | 959   | 1035  | 1008  | 769   | 495   | 182   | 53    |
|                                                                        | CDD65       | 2528   | 0     | 4     | 29    | 111   | 312   | 509   | 570   | 543   | 325   | 116   | 9     | 0     |
|                                                                        | CDH74       | 31343  | 24    | 124   | 519   | 1519  | 3828  | 6414  | 7335  | 6721  | 3430  | 1242  | 161   | 26    |
|                                                                        | CDH80       | 16394  | 1     | 22    | 144   | 606   | 1996  | 3635  | 4159  | 3723  | 1617  | 464   | 27    | 0     |
| (14) Wind                                                              | WSAvg       | 10.9   | 10.1  | 11.0  | 11.8  | 12.8  | 12.5  | 12.4  | 10.7  | 9.7   | 9.6   | 10.3  | 9.8   | 9.8   |
| (15) Precipitation                                                     | PrecAvg     | 14.4   | 0.6   | 0.6   | 0.6   | 0.7   | 1.8   | 1.7   | 1.7   | 1.9   | 2.2   | 1.5   | 0.7   | 0.6   |
|                                                                        | PrecMax     | 32.1   | 2.5   | 2.6   | 2.9   | 2.5   | 7.6   | 4.0   | 8.5   | 6.5   | 9.7   | 7.4   | 5.5   | 3.3   |
|                                                                        | PrecMin     | 5.1    | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   |
|                                                                        | PrecStd     | 5.3    | 0.6   | 0.6   | 0.6   | 0.7   | 1.5   | 1.1   | 2.0   | 1.5   | 1.9   | 1.7   | 0.9   | 0.7   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 77.3  | 83.4  | 88.6  | 94.0  | 102.2 | 106.5 | 103.6 | 102.8 | 98.8  | 93.1  | 83.8  | 77.8  |
|                                                                        |             | MCWB   | 52.2  | 54.2  | 56.2  | 57.9  | 63.9  | 66.5  | 68.3  | 68.9  | 66.6  | 63.3  | 57.5  | 52.5  |
|                                                                        | 2%          | DB     | 72.0  | 77.8  | 84.0  | 90.2  | 98.2  | 102.2 | 100.2 | 99.6  | 95.1  | 89.0  | 78.3  | 71.9  |
|                                                                        |             | MCWB   | 49.9  | 52.3  | 55.5  | 58.1  | 62.6  | 66.5  | 68.5  | 68.7  | 66.8  | 63.3  | 56.9  | 51.5  |
|                                                                        | 5%          | DB     | 67.4  | 73.3  | 80.1  | 87.1  | 94.7  | 98.8  | 98.0  | 97.3  | 92.3  | 85.5  | 74.6  | 66.8  |
|                                                                        |             | MCWB   | 48.2  | 51.4  | 54.5  | 57.6  | 62.3  | 67.3  | 68.6  | 68.8  | 66.9  | 62.6  | 55.3  | 49.1  |
|                                                                        | 10%         | DB     | 62.2  | 68.2  | 75.7  | 83.4  | 91.0  | 95.6  | 95.8  | 94.9  | 89.3  | 81.4  | 70.4  | 62.0  |
|                                                                        |             | MCWB   | 46.7  | 49.5  | 53.7  | 57.4  | 62.6  | 67.7  | 68.6  | 68.8  | 66.8  | 61.6  | 54.2  | 48.2  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 57.1  | 59.9  | 63.8  | 67.2  | 71.9  | 74.3  | 74.0  | 73.9  | 72.7  | 71.2  | 64.7  | 59.2  |
|                                                                        |             | MCDB   | 66.0  | 69.8  | 73.8  | 79.5  | 85.0  | 90.4  | 88.7  | 89.6  | 85.5  | 82.8  | 71.3  | 65.2  |
|                                                                        | 2%          | WB     | 53.3  | 57.0  | 61.5  | 65.3  | 70.2  | 73.0  | 72.6  | 72.5  | 71.4  | 69.0  | 61.3  | 55.0  |
|                                                                        |             | MCDB   | 63.4  | 67.0  | 71.2  | 76.0  | 83.1  | 88.7  | 88.2  | 88.7  | 83.8  | 79.0  | 69.9  | 64.5  |
|                                                                        | 5%          | WB     | 50.4  | 54.1  | 59.5  | 63.6  | 68.8  | 71.8  | 71.7  | 71.6  | 70.5  | 67.4  | 59.2  | 51.6  |
|                                                                        |             | MCDB   | 63.6  | 67.7  | 70.3  | 74.4  | 81.9  | 87.3  | 87.4  | 87.9  | 83.0  | 76.9  | 68.5  | 63.6  |
|                                                                        | 10%         | WB     | 47.7  | 51.4  | 57.0  | 61.6  | 67.5  | 70.9  | 71.0  | 70.9  | 69.4  | 65.4  | 56.8  | 48.7  |
|                                                                        |             | MCDB   | 61.1  | 65.6  | 69.9  | 74.7  | 81.1  | 86.3  | 86.8  | 87.0  | 82.1  | 74.9  | 67.0  | 61.1  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 26.1  | 26.7  | 27.3  | 27.9  | 25.7  | 24.5  | 22.7  | 22.6  | 22.5  | 24.3  | 25.2  | 25.2  |
|                                                                        |             | MCDBR  | 33.9  | 34.3  | 32.6  | 31.7  | 30.2  | 28.5  | 25.3  | 25.3  | 26.5  | 28.8  | 30.7  | 33.1  |
|                                                                        |             | MCWBR  | 16.8  | 15.9  | 13.3  | 12.6  | 11.1  | 8.3   | 7.2   | 6.8   | 8.2   | 10.9  | 14.0  | 16.9  |
|                                                                        | 5% WB       | MCDBR  | 29.2  | 28.4  | 26.4  | 25.8  | 25.3  | 23.6  | 21.9  | 22.1  | 20.5  | 21.3  | 23.7  | 27.7  |
|                                                                        |             | MCWBR  | 15.8  | 15.8  | 13.2  | 13.0  | 11.0  | 7.8   | 6.5   | 6.5   | 6.9   | 9.7   | 13.2  | 15.9  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.265 | 0.276 | 0.295 | 0.331 | 0.348 | 0.378 | 0.396 | 0.373 | 0.353 | 0.314 | 0.287 | 0.270 |
|                                                                        | taud        |        | 2.578 | 2.548 | 2.507 | 2.400 | 2.415 | 2.379 | 2.374 | 2.413 | 2.437 | 2.530 | 2.540 | 2.564 |
|                                                                        | Ebn at Noon |        | 310   | 315   | 314   | 304   | 297   | 287   | 281   | 287   | 290   | 297   | 298   | 302   |
|                                                                        | Edh at Noon |        | 25    | 29    | 32    | 37    | 37    | 39    | 39    | 37    | 34    | 29    | 26    | 24    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1101  | 1363  | 1764  | 2157  | 2288  | 2412  | 2288  | 2122  | 1770  | 1451  | 1155  | 984   |
|                                                                        | RadStd      |        | 107   | 142   | 126   | 134   | 159   | 111   | 125   | 93    | 120   | 160   | 121   | 88    |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | +0.85 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +295  | +216  |
| (47) Regional (17 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page